Global Financial Spillovers to the NIFTY 50: Evidence from S&P 500 , Crude Oil, and VIX

Presented by Gautam Prasad

Research Objective

How do global financial market shocks transmit to NIFTY 50, and is NIFTY 50 a net transmitter or receiver of cross-market spillovers?

Data Description

Assets - NIFTY 50(^NSEI), S&P 500 (^GSPC), VIX (^VIX), WTI Crude Oil(CL=F)
Source - Yahoo
Timeline - 01/01/2013 to 01/01/2025

Methodology Explanation

1. Data Transformation

- Daily prices converted to log-returns.
- Ensures stationarity and comparability across markets.

2. VAR Model

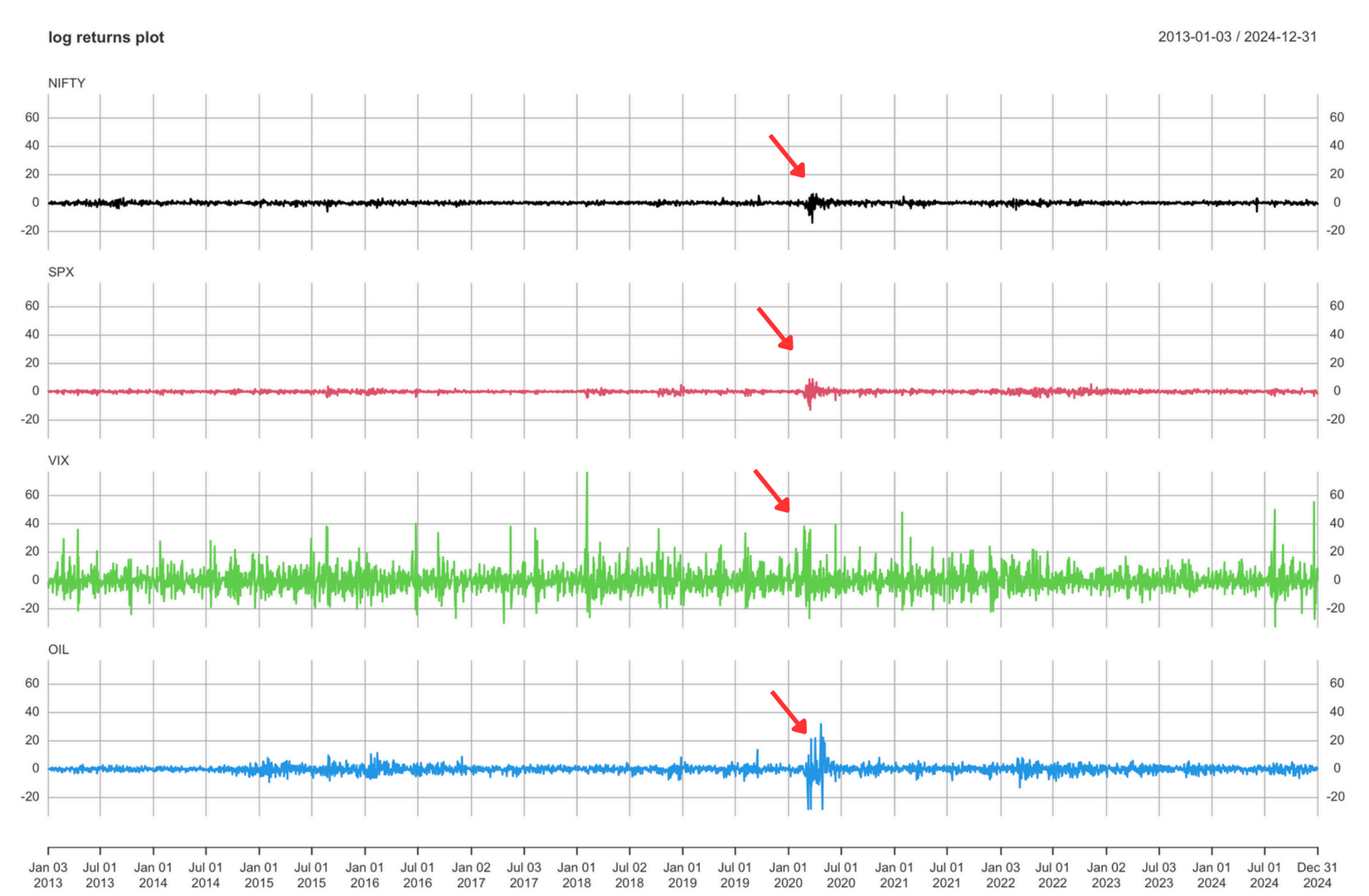
- A Vector Autoregression (VAR) is estimated on:
- NIFTY 50, S&P 500, VIX, Crude Oil
- Each variable depends on its own lags and the lags of all others.
- Lag length selected using Schwarz Information Criterion.

Methodology Explanation

3. Diebold–Yilmaz Spillovers (DY12)
   - Uses generalized forecast error variance decomposition.
   - Measures how much uncertainty in one market is explained by shocks in others.
   - Produces:
     - Total Connectedness Index (TCI),
     - Directional spillovers (TO / FROM),
     - Net spillovers,
     - Pairwise spillovers.

4. Dynamic Connectedness
   - VAR is re-estimated using a rolling window (180 days).
   - Spillovers are computed for each window.

log return plot



- All four series exhibit higher spikes during COVID-19, indicating strong cross-market shock transmission during crisis periods.

Estimation results

Var Model

```
Estimation results for equation NIFTY:
======================================
NIFTY = NIFTY.l1 + SPX.l1 + VIX.l1 + OIL.l1 + NIFTY.l2 + SPX.l2 + VIX.l2 + OIL.l2 + const

          Estimate Std. Error t value Pr(>|t|)
NIFTY.l1 -0.093355   0.019019  -4.908 9.66e-07 ***
SPX.l1    0.236788   0.025310   9.356  < 2e-16 ***
VIX.l1   -0.006471   0.003275  -1.976  0.04823 *
OIL.l1    0.023791   0.006507   3.656  0.00026 ***
NIFTY.l2  0.005476   0.018066   0.303  0.76182
SPX.l2    0.081332   0.025647   3.171  0.00153 **
VIX.l2   -0.002749   0.003291  -0.835  0.40359
OIL.l2   -0.026940   0.006517  -4.134 3.66e-05 ***
const     0.033731   0.017191   1.962  0.04984 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

- S&P 500 & NIFTY 50 → Positive association
- VIX & NIFTY 50 → Negative association
- Crude Oil & NIFTY → Mixed association

```
Roots of the characteristic polynomial:
0.274 0.274 0.2669 0.2591 0.2591 0.1391 0.1391 0.111
```

VAR(2) estimated on daily log returns . Model stability confirmed (All characteristic roots < 1).

Static Spillover Results (Diebold–Yilmaz Framework 2012)

```
> print(sp12)
The spillover table has no frequency bands, standard Diebold & Yilmaz.


|       | NIFTY|   SPX|   VIX|   OIL|  FROM|
|:-----|-----:|-----:|-----:|-----:|-----:|
|NIFTY  | 74.77| 14.51|  8.65|  2.07|  6.31|
|SPX    |  8.11| 57.97| 31.11|  2.80| 10.51|
|VIX    |  3.86| 33.43| 60.67|  2.05|  9.83|
|OIL    |  1.03|  4.81|  3.27| 90.89|  2.28|
|TO     |  3.25| 13.19| 10.76|  1.73| 28.93|
>
```

```
> net(sp12)
[[1]]
     NIFTY        SPX        VIX        OIL
-3.0569593  2.6788804  0.9237168 -0.5456379
```

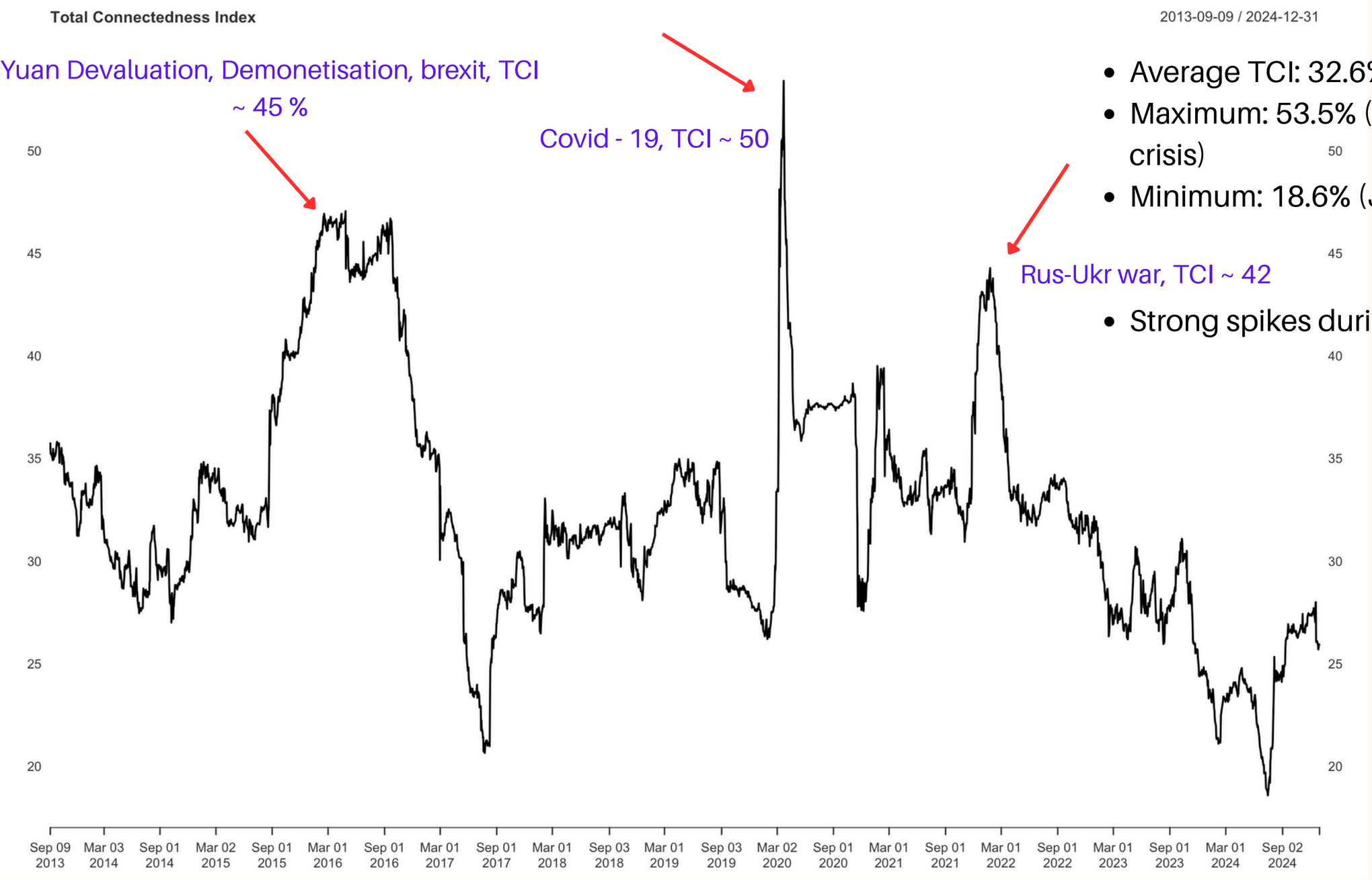
- Total Connectedness Index (TCI): 28.9%

Net Spillovers:

- NIFTY 50 : −3.06 (net receiver)
- S&P 500: +2.68 (net transmitter)
- VIX: +0.92 (net transmitter)
- Crude Oil: −0.55 (net receiver)

- About 29% of market fluctuations is explained by cross-market spillovers.
- The US market and global risk sentiment are the main sources of shocks.

Dynamic Spillover - TCI



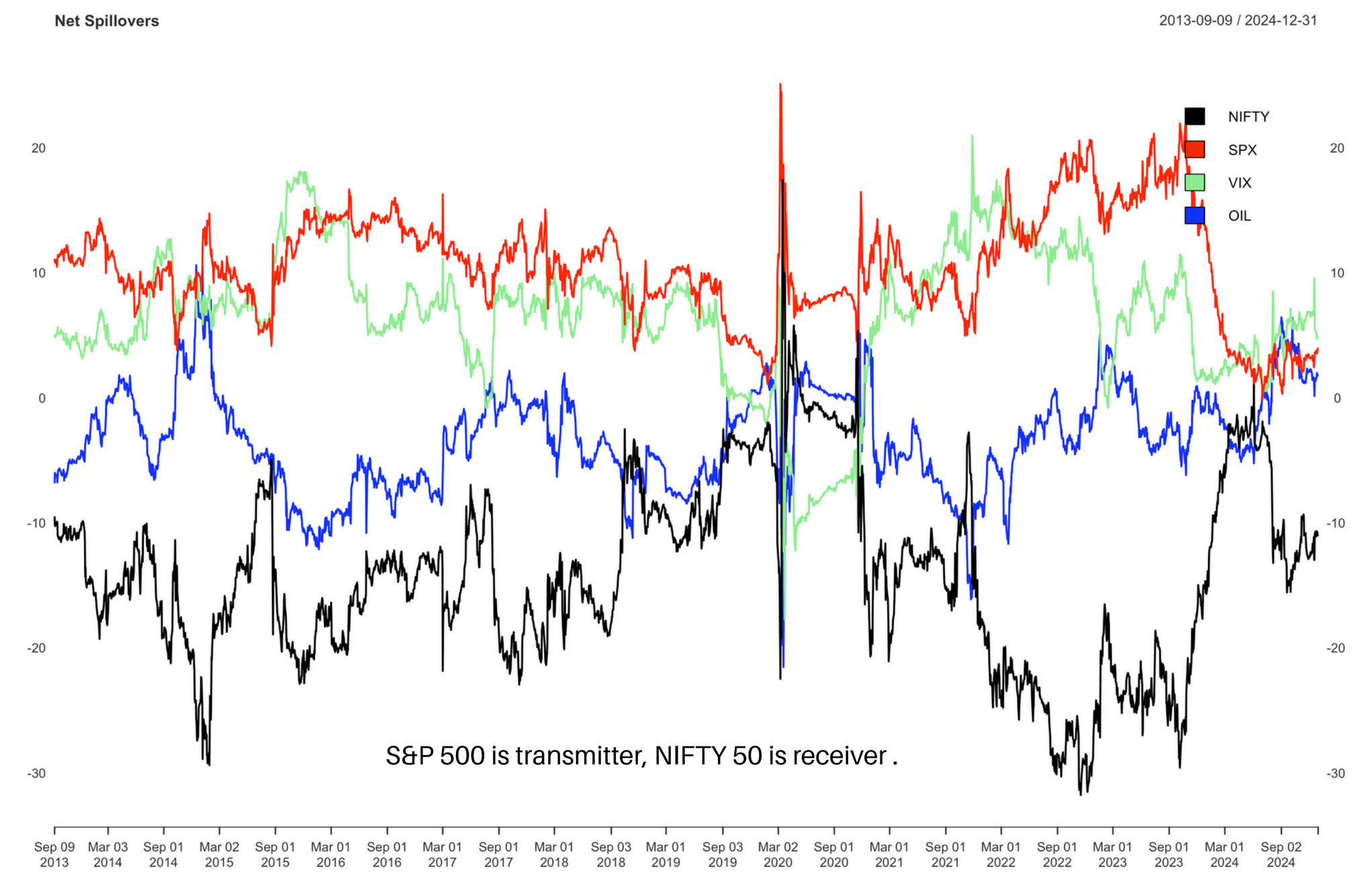
- Average TCI: 32.6%
- Maximum: 53.5% (March 2020 – COVID crisis)
- Minimum: 18.6% (July 2024 – stable period)

- Strong spikes during global crisis periods

Average spillover to nifty during the spikes

```
> crisis_table
   Period      SPX       VIX      OIL
1 2015-16 15.50426 17.295462 2.370540
2   COVID 19.25778  7.251076 8.064459
3    2022 17.78832 16.374250 2.042714
```

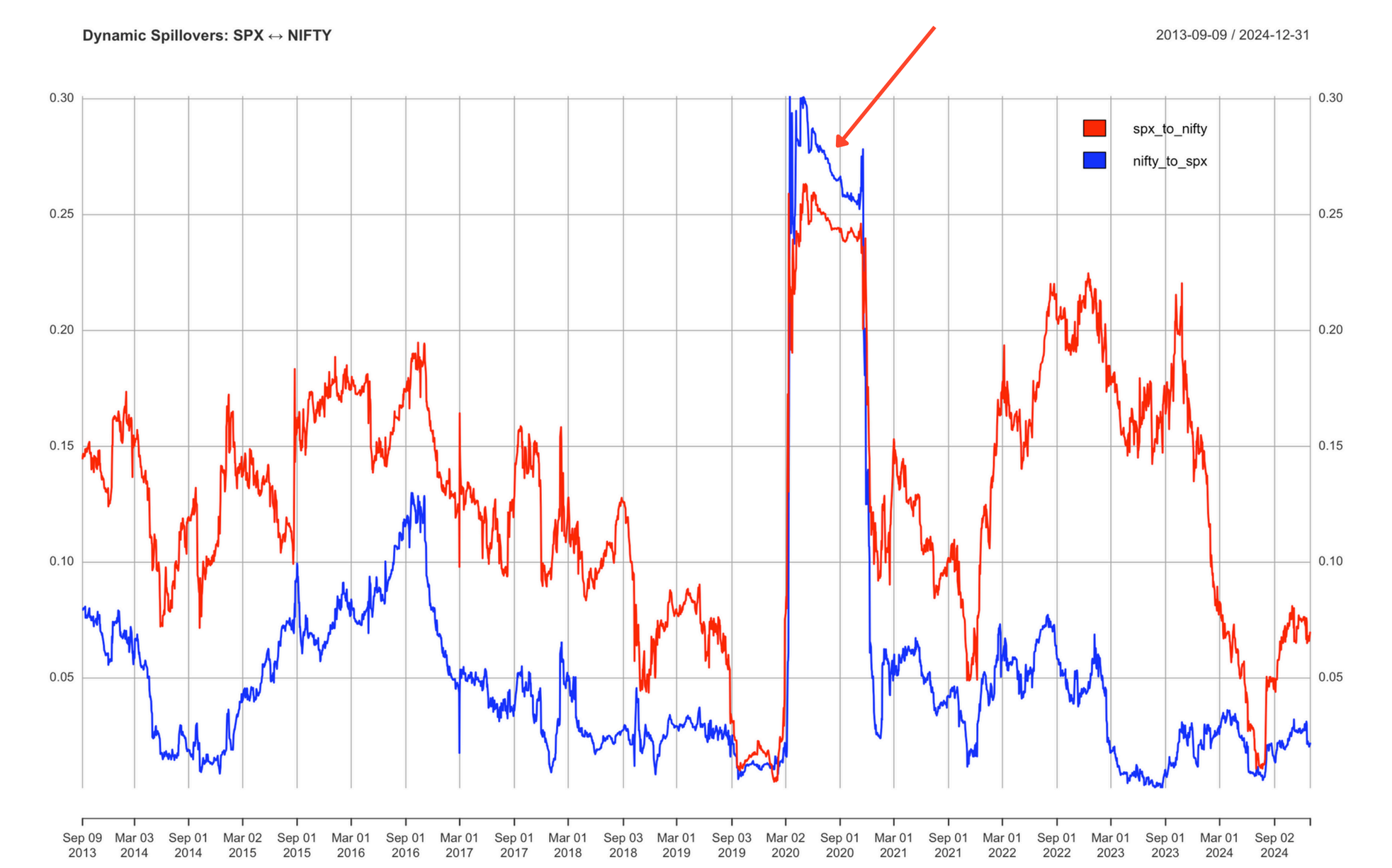
NET Spillover



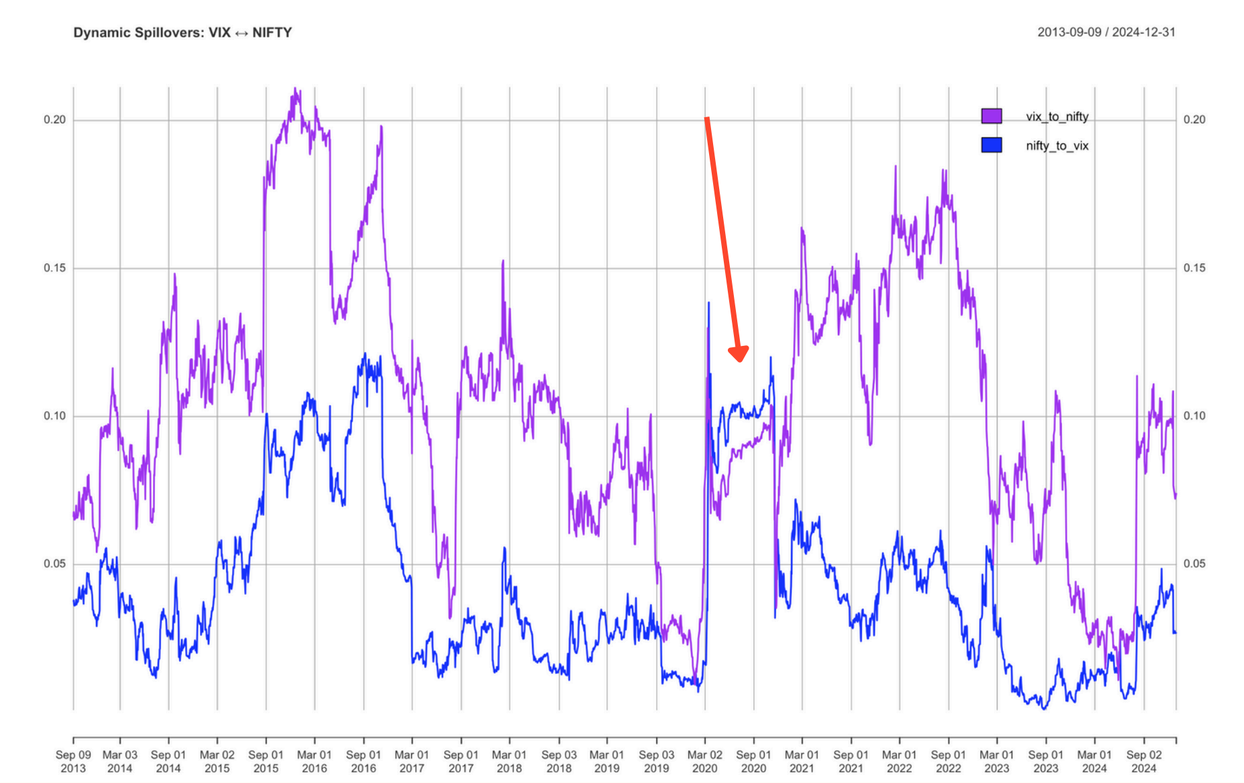
S&P 500 is transmitter, NIFTY 50 is receiver .

Pairwise spillover - Nifty & Spx

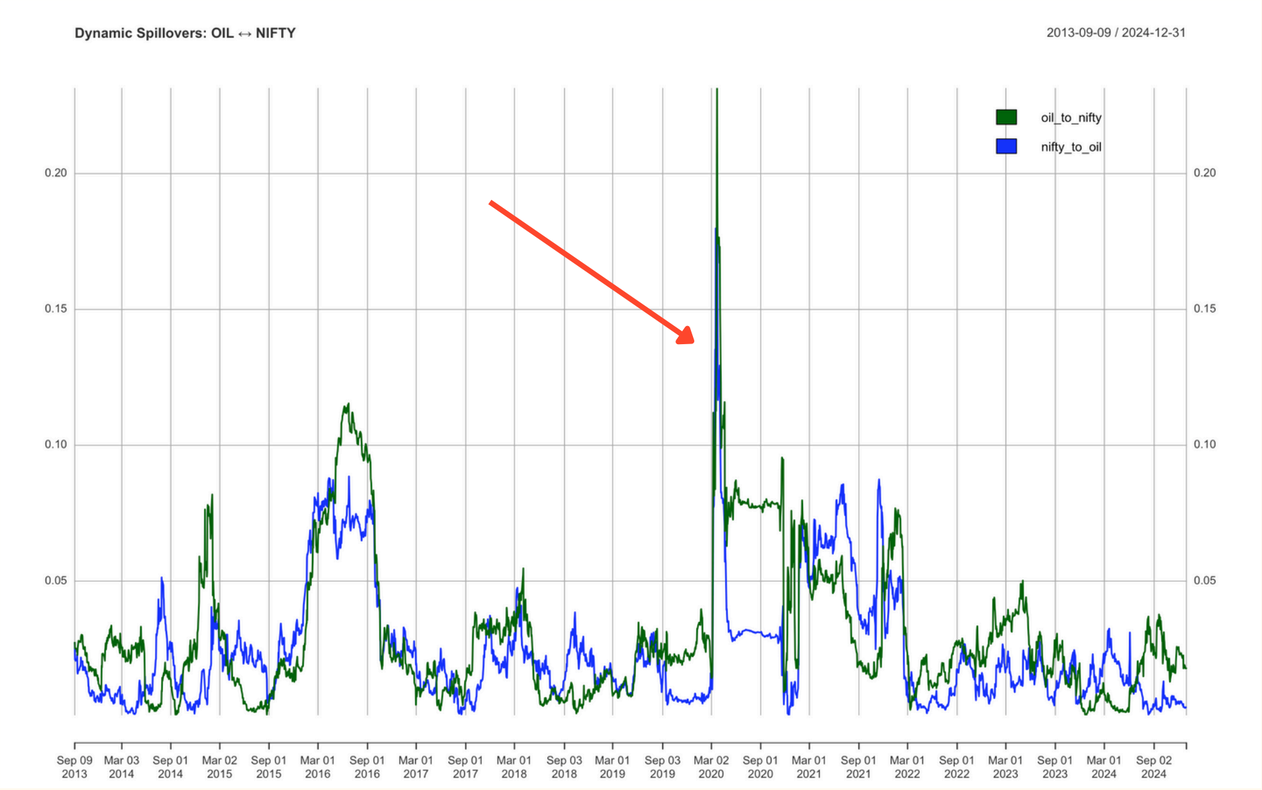
During COVID, markets were reacting simultaneously to a common global shock



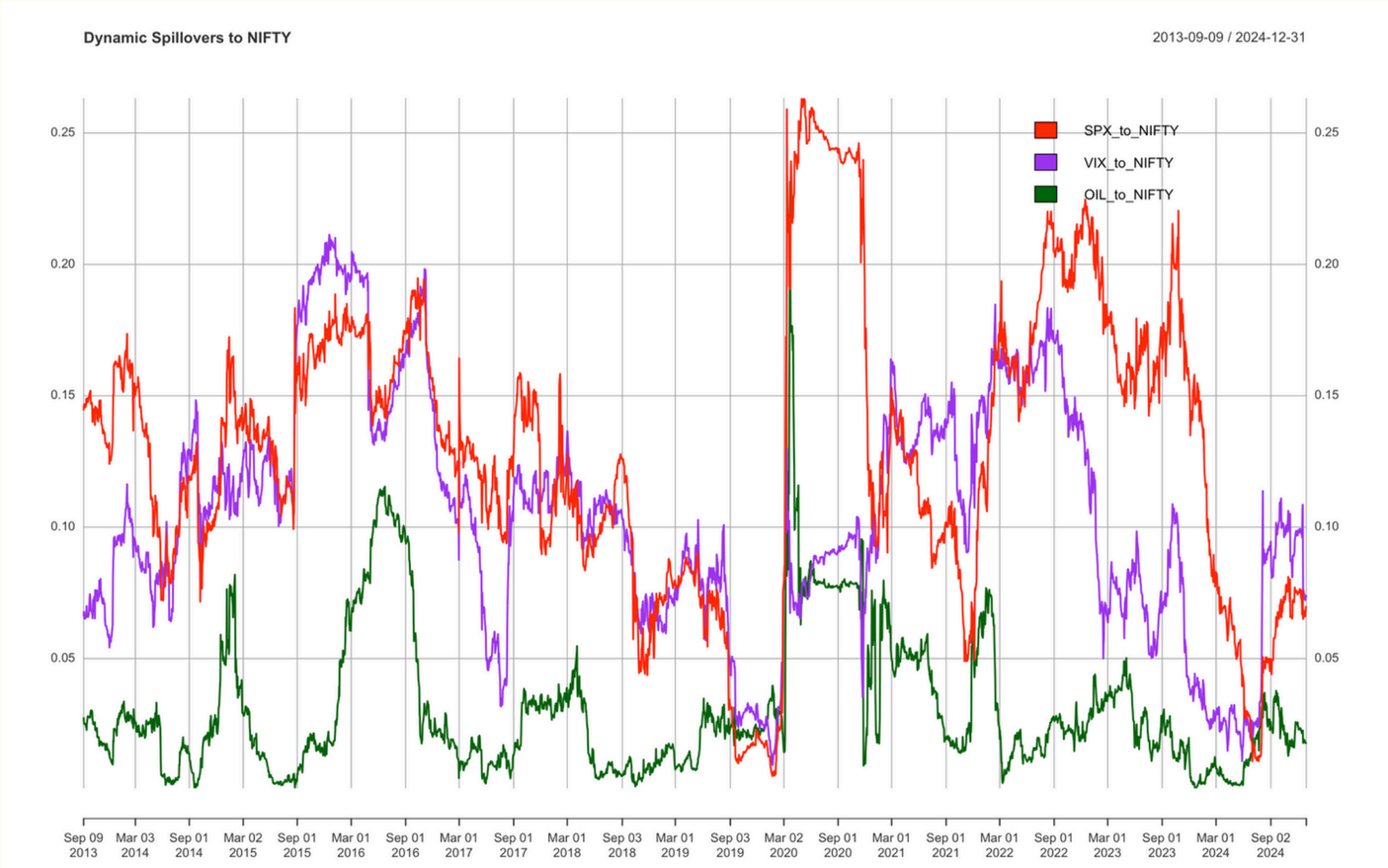
Pairwise spillover - NIFTY 50 & VIX

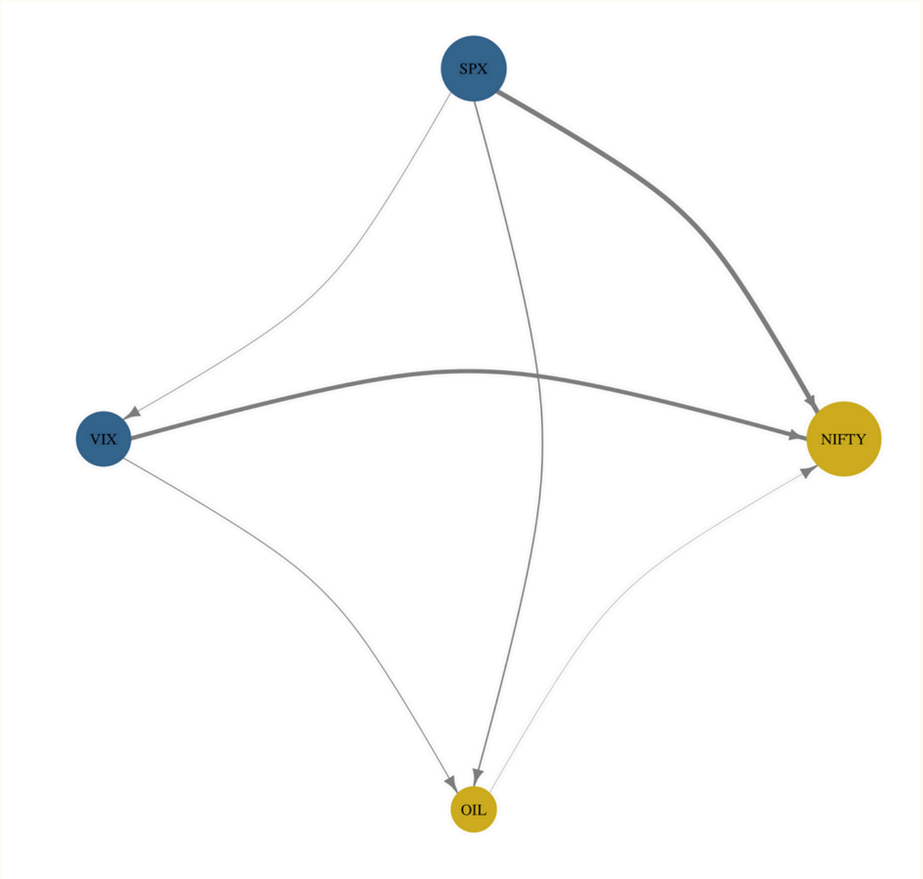


Pairwise spillover - NIFTY 50 & Crude Oil



Conclusion





- NIFTY 50 is primarily a net receiver of global return spillovers.
- Spillovers originate mainly from the U.S. equity market (S&P 500) and changes in global risk sentiment captured by the VIX.
- Spillover intensity increases during global crisis periods
- Overall, NIFTY 50 Index is financially integrated but structurally vulnerable to global shocks

Thank You